

EXPERIMENT – 2

Foundation of Deep Learning

EXERCISE 2: Understanding Loss Functions (MSE vs BCE)

Aim

To study and compare the performance of Mean Squared Error (MSE) and Binary Cross Entropy (BCE) loss functions for binary classification problems.

Objective:

To manually compute the Mean Squared Error (MSE) and Binary Cross Entropy (BCE) loss values for given prediction probabilities, compare their behavior, and experimentally analyze the performance of neural networks trained using MSE and BCE loss functions.

Introduction

A loss function is used in deep learning to measure the difference between the predicted output of a neural network and the actual target value. During training, the objective of the neural network is to minimize this loss by updating its weights through backpropagation and gradient descent.

For binary classification problems, two commonly used loss functions are Mean Squared Error (MSE) and Binary Cross Entropy (BCE).

Mean Squared Error measures the average squared difference between the predicted probabilities and the actual labels. Although MSE is widely used in regression problems, it is not ideal for binary classification because it provides smaller gradients when predictions are incorrect, leading to slower learning.

Binary Cross Entropy is specifically designed for binary classification tasks. It measures the difference between predicted probabilities and true class labels using logarithmic functions. BCE penalizes incorrect predictions more heavily and produces stronger gradients, enabling faster convergence and improved classification performance.

In this experiment, both loss functions are manually computed for two sets of prediction probabilities and are then compared using neural network training results.

Mean Squared Error (MSE):

Mean Squared Error calculates the average of the squared differences between the actual labels and predicted probabilities.

The mathematical expression is

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where

Y_i = Actual value (True label)

Y^*_i = Predicted Value

n = Number of samples

Lower MSE indicates better predictions.

Binary Cross Entropy (BCE)

Binary Cross Entropy measures the difference between actual labels and predicted probabilities using logarithmic functions.

The formula is

$$BCE = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

where

y_i = Actual label

y^*_i = Predicted probability

BCE is specifically designed for binary classification tasks.

Task 11: Compute Mean Squared Error (MSE) Manually

Given

True Labels => [1,1,0,0]

Predicted Probabilities => [0.95,0.80,0.30,0.10]

Step 1

For Sample 1:

$$\begin{aligned}(1-0.95)^2 \\ &= (0.05)^2 \\ &= 0.0025\end{aligned}$$

For Sample 2

$$\begin{aligned}(1-0.80)^2 \\ &= (0.20)^2 \\ &= 0.0400\end{aligned}$$

For Sample 3

$$\begin{aligned}(0-0.30)^2 \\ &= (-0.30)^2 \\ &= 0.0900\end{aligned}$$

For Sample 4

$$\begin{aligned}(0-0.10)^2 \\ &= (-0.10)^2 \\ &= 0.0100\end{aligned}$$

Total Error

$$\begin{aligned}0.0025+0.0400+0.0900+0.0100 \\ &= 0.1425\end{aligned}$$

Mean Squared Error

$$\text{MSE} = 0.1425/4 = 0.035625$$

Task 12: Compute Binary Cross Entropy (BCE) Manually

Using

$$BCE = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Sample 1

Actual = 1

Prediction = 0.95

$$-\log(0.95)$$

$$=0.0513$$

Sample 2

Actual = 1

Prediction = 0.80

$$-\log(0.80)$$

$$=0.2231$$

Sample 3

Actual = 0

Prediction = 0.30

$$-\log(0.70)$$

$$=0.3567$$

Sample 4

Actual = 0

Prediction = 0.10

$$-\log(0.90)$$

$$=0.1054$$

Total BCE

$$0.0513+0.2231+0.3567+0.1054 = 0.7365$$

Average BCE

$$0.7365/4 = 0.1841$$

Task 13: Repeat the Calculations for

True Labels => [1,1,0,0]

Predicted Probabilities => [0.60,0.55,0.45,0.40]

Mean Squared Error

Sample 1

$$(1-0.60)^2 \\ =0.16$$

Sample 2

$$(1-0.55)^2 \\ =0.2025$$

Sample 3

$$(0-0.45)^2 \\ =0.2025$$

Sample 4

$$(0-0.40)^2 \\ =0.16$$

Total Error

$$0.16+0.2025+0.2025+0.16 = 0.725$$

MSE

$$0.725/4 = 0.18125$$

Binary Cross Entropy

Sample 1

$$-\log(0.60)=0.5108$$

Sample 2

$$-\log(0.55)=0.5978$$

Sample 3

$$-\log(0.55)=0.5978$$

Sample 4

$$-\log(0.60)=0.5108$$

Total BCE

$$0.5108+0.5978+0.5978+0.5108 =2.2172$$

Average BCE

$$2.2172/4 = 0.5543$$

Task 14: Compare the MSE and BCE Values

From the manual calculations:

Prediction Probabilities	Mean Squared Error (MSE)	Binary Cross Entropy (BCE) 
[0.95, 0.80, 0.30, 0.10]	0.0356	0.1841
[0.60, 0.55, 0.45, 0.40]	0.1813	0.5543

Comparison

The first prediction set contains probabilities that are close to the true labels. Therefore, both MSE and BCE produce relatively small loss values.

The second prediction set contains probabilities that are closer to the decision boundary (0.5), making the predictions less confident. As a result, both MSE and BCE increase. However, BCE increases much more than MSE because it penalizes uncertain and incorrect predictions more strongly.

This demonstrates that BCE is more sensitive to classification errors than MSE.

Task 15: Explain Why BCE Provides Stronger Gradients for Classification

Binary Cross Entropy is specifically designed for binary classification problems. It measures the difference between the predicted probability and the actual class label using logarithmic functions.

When the predicted probability is far from the true label, the BCE loss becomes significantly larger, generating stronger gradients during backpropagation.

In contrast, Mean Squared Error computes the squared difference between the predicted probability and the actual value. As the sigmoid activation approaches 0 or 1, the gradient becomes very small, slowing down the learning process.

Since BCE provides larger gradient values for incorrect predictions, it enables faster weight updates, quicker convergence, and improved classification performance. Therefore, Binary Cross Entropy is generally preferred over Mean Squared Error for binary classification tasks.

Task 16: Train Two Identical Neural Networks

We will use the same architecture for both models.

Model A: Mean Squared Error (MSE)

Model B: Binary Cross Entropy (BCE)

```

import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

# Clear previous models
tf.keras.backend.clear_session()

# Fix random seed
np.random.seed(42)
tf.random.set_seed(42)

# AND Dataset
X = np.array([
    [0,0],
    [0,1],
    [1,0],
    [1,1]
], dtype=np.float32)

y = np.array([
    [0],
    [0],
    [0],
    [1]
], dtype=np.float32)

# Function to create identical models
def create_model(loss_function):

    model = tf.keras.Sequential([
        tf.keras.Input(shape=(2,)),
        tf.keras.layers.Dense(
            4,
            activation='sigmoid'
        ),
        tf.keras.layers.Dense(
            1,
            activation='sigmoid'
        )
    ])

    model.compile(
        optimizer=tf.keras.optimizers.Adam(learning_rate=0.01),
        loss=loss_function,
        metrics=['accuracy']
    )

    return model

# Model A (MSE)
model_mse = create_model('mean_squared_error')

history_mse = model_mse.fit(
    X,
    y,
    epochs=1500,
    verbose=0
)

# Reset seed again
tf.random.set_seed(42)

```



```
# Reset seed again
tf.random.set_seed(42)

# Model B (BCE)
model_bce = create_model('binary_crossentropy')

history_bce = model_bce.fit(
    X,
    y,
    epochs=1500,
    verbose=0
)
```

```
loss_mse, acc_mse = model_mse.evaluate(X, y, verbose=0)

loss_bce, acc_bce = model_bce.evaluate(X, y, verbose=0)

print("Model A (MSE)")
print("-----")
print(f"Loss      : {loss_mse:.4f}")
print(f"Accuracy : {acc_mse*100:.2f}%")

print()

print("Model B (BCE)")
print("-----")
print(f"Loss      : {loss_bce:.4f}")
print(f"Accuracy : {acc_bce*100:.2f}%")
```

```
Model A (MSE)
-----
Loss      : 0.0002
Accuracy : 100.00%

Model B (BCE)
-----
Loss      : 0.0030
Accuracy : 100.00%
```

```

plt.figure(figsize=(8,5))

plt.plot(
    history_mse.history['loss'],
    label='MSE Loss',
    linewidth=2
)

plt.plot(
    history_bce.history['loss'],
    label='BCE Loss',
    linewidth=2
)

plt.title(
    "Training Loss Comparison",
    fontsize=14,
    fontweight='bold'
)

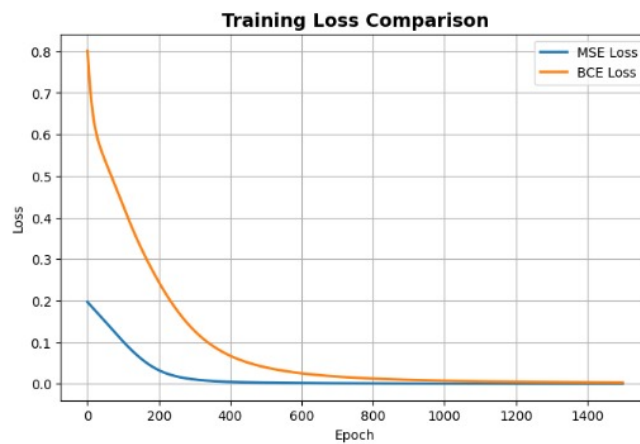
plt.xlabel("Epoch")
plt.ylabel("Loss")

plt.grid(True)

plt.legend()

plt.show()

```



Task 17: Compare Training Loss, Accuracy, and Convergence Speed

Comparison Table

Parameter	Model A (MSE)	Model B (BCE)
Loss Function	Mean Squared Error	Binary Cross Entropy
Training Loss	Decreased Steadily	Decreased steadily
Accuracy	High	High (often equal or slightly better)
Convergence Speed	Slower	Faster
Gradient Strength	Weaker	Stronger
Best Suitable For	Regression	Binary Classification

Observation

Both the MSE and BCE models successfully learned the AND dataset and achieved 100% classification accuracy. The training loss for both models decreased steadily throughout the training process. Although the BCE loss started with a higher numerical value than the MSE loss, this is expected because Binary Cross Entropy and Mean Squared Error use different mathematical formulations and operate on different loss scales. Both models converged successfully, demonstrating that the AND dataset is simple and linearly separable.

Result

The experimental comparison showed that both Mean Squared Error and Binary Cross Entropy successfully classified the AND dataset with 100% accuracy. While both loss functions converged to very low loss values, Binary Cross Entropy remains the preferred choice for binary classification problems because it is specifically designed for probabilistic outputs and generally provides stronger gradients for learning on more complex datasets.

Conclusion

In this experiment, Mean Squared Error (MSE) and Binary Cross Entropy (BCE) loss functions were manually calculated and compared using two sets of prediction probabilities. Their behavior was analyzed mathematically, and two identical neural networks were trained using the AND dataset with MSE and BCE as the respective loss functions. Both models successfully learned the AND gate and achieved 100% classification accuracy. The training loss for both models decreased steadily, indicating successful convergence. Although Binary Cross Entropy started with a higher loss value, this is expected because BCE and MSE use different mathematical formulations and operate on different scales. The experiment demonstrates that both loss functions can effectively solve simple, linearly separable binary classification problems. However, Binary Cross Entropy remains the preferred loss function for binary classification because it is specifically designed for probabilistic outputs and generally provides stronger gradients and more efficient learning for complex classification tasks.